

GUEST SATISFACTION ANALYSIS OF AIRBNB LISTINGS IN ITALY

Technical Report

Technical Summary: This project analysed Airbnb listings in Italy to identify the factors influencing guest satisfaction and provide evidence-based insights that help tourists make informed accommodation choices. Microsoft Excel was used for data cleaning, while Microsoft Power BI was used for data modelling, analysis, dashboard development, and visualisation.

The analysis focused on guest experience factors, pricing, property characteristics, host characteristics, and location to determine how each contributes to guest satisfaction.

1. Introduction & Business Context

Italy is one of the world's most visited tourist destinations, offering thousands of Airbnb listings across multiple cities. With such a wide range of accommodation options available, tourists often find it difficult to determine which listings provide the best overall experience.

Guest satisfaction is influenced by several factors, including cleanliness, value for money, host characteristics, property characteristics, pricing, and location. Understanding the relationship between these factors enables tourists to make informed accommodation choices based on data rather than assumptions.

Business Questions

- A. Which aspects of the guest experience have the strongest relationship with overall guest satisfaction?
- B. How does price influence guest satisfaction and value for money?
- C. How do host characteristics influence guest satisfaction?
- D. Which property types and property sizes have the greatest influence on guest satisfaction?
- E. Which neighbourhoods and cities offer the best value for money?

2. Data Overview

Data Source: The dataset was obtained from Kaggle. According to the dataset documentation, the data was originally scraped from Inside Airbnb, an independent platform that provides publicly available Airbnb listing data for research and analysis.

Dataset Contents.

The dataset contains; Listing ID, City, Neighbourhood, Property Type, Bedrooms
Beds, Bathrooms, Maximum Guests, Price
Host Since, Host Status, Superhost Status
Rating Score, Accuracy Score, Cleanliness Score, Check-in Score, Communication Score,
Location Score, Value for Money Score, Reviews, Last Year Reviews.

3. Detailed Methodology

3.1 Data Cleaning (Microsoft Excel)

The dataset underwent preprocessing before analysis.

- Cleaning activities included:
- Removal of duplicate records.
- Standardisation of data formats.
- Creation of the Price Flag column.
- Identification of abnormal price values.

Price Flag

A total of 650 listings contained unusually high prices above **€2,000**, with some reaching **€95,195**.

These records were flagged as "Check this price" because they were unusually high relative to similar listings and may represent data entry errors, such as monthly prices being entered as nightly prices.

Since the true values could not be verified, these records remained in the dataset but were excluded from price-related analyses.

Documentation of listings with Overall Rating equal to 0.00.

- Overall Rating = 0.00
- Twenty-three listings contained: Last Year Reviews = 0 (or near-zero reviews)

The dataset documentation did not define whether these values represented missing ratings or genuine ratings.

These records remained in the dataset but were excluded from calculations involving Overall Rating.

3.2 Data Analysis (Microsoft Power BI)

i. Business Question A& D– Guest Experience Drivers and Property Characteristics

The cleaned dataset was imported into Power BI.

Several DAX measures were created to calculate the average values of guest experience metrics.

Examples include:

Average Rating =
AVERAGE('Italy file'[Rating score])

The same approach was applied to calculate:

- Average Cleanliness
- Average Communication
- Average Accuracy
- Average Check-in
- Average Location
- Average Value for Money

A calculated column named Property Size was created to categorise listings based on bedroom numbers.

Property Size =
SWITCH(TRUE(),
'Italy file'[Bedrooms Number]=1,"Small",
'Italy file'[Bedrooms Number]<3,"Medium",
'Italy file'[Bedrooms Number]<=5,"Large",
"Very Large")

Properties were grouped as:

- Small (1 bedroom)
- Medium (2–3 bedrooms)
- Large (4–5 bedrooms)
- Very Large (6–10 bedrooms)

To identify the strongest drivers of guest satisfaction, Power BI's Key Influencers visual was used.

The visual analysed:

Rating Score was inputted into (Analyse field) against the six other metrics(Value for Money, Cleanliness, Accuracy, Communication, Check-in,Location) inputted into (Explain By fields).

The Key Influencers visual applies built-in AI techniques, including correlation and regression analysis, to determine which variables have the greatest influence on the analysed metric.

The visual showed that Value for Money had the strongest influence on guest satisfaction.

A scatter chart was also developed with:

X-axis = Value for Money

Y-axis = Rating Score

Listing ID was converted to text and added to the visual so each listing appeared as an individual observation.

The upward trend of the scatter plot confirmed a strong positive relationship between Value for Money and Overall Rating.

Property Size and Property Type were analysed using clustered column charts to compare average guest ratings across categories.

ii. Business Question B – Pricing Analysis

A calculated column named Price Band was created to classify listings into pricing groups. This made it easier to analyze and visualize the relationship between price and guest satisfaction, especially because the dataset contained a large range of prices.

Measures were created to summarize key performance indicators in the dataset. The Average Price and Median Price helped understand the pricing distribution, the Average Rating measured overall guest satisfaction, the Total Listings measure provided context for the dataset size, and the Average Price of Highly Rated Listings helped assess whether highly rated accommodations are associated with higher prices.

KPI cards were used to summarize the key metrics, a clustered column chart to compare guest satisfaction across price bands, and a scatter chart with a trend line to examine the relationship between price and guest ratings. Together, these visuals answer the main business question of whether higher prices lead to better guest satisfaction.

We found out that paying a higher price may lead to slightly better guest satisfaction, but the difference is small.

This suggests that price alone does not determine guest satisfaction.

To provide additional insight, we calculated the average price of highly rated listings to determine whether guests have to pay significantly more for highly rated accommodations. The analysis showed that highly rated listings have an average price of **€178.50**, compared with the overall average price of **€173.16**. The small difference of **€5.34** suggests that guests can find highly rated accommodations without paying significantly more than the market average.

Measures created included:

- Average Price
- Median Price
- Average Rating
- Total Listings
- Average Price of Highly Rated Listings

Visuals developed:

- KPI Cards
- Clustered Column Chart

- Scatter Plot with Trend Line

These visuals examined the relationship between pricing and guest satisfaction.

iii. Business Question C & E – Host Characteristics and Location Analysis

A calculated column named Years of Experience was created using the DATEDIF function to subtract the host since from the date of scraping

Created DAX measures to split host vs. superhost ratings and tenure:

Host rating =
`CALCULATE(AVERAGE(Table1[Rating score]), Table1[Host is superhost]="Host"),`

Superhost rating =
`CALCULATE(AVERAGE(Table1[Rating score]), Table1[Host is superhost]= "Superhost")`

Host years =
`CALCULATE(AVERAGE(Table1[Years of experience]), Table1[Host is superhost]= "Host")`

Superhost years =
`CALCULATE(AVERAGE(Table1[Years of experience]), Table1[Host is superhost]= "Superhost")`

Built the pie and donut charts from those measures

Built the city-level bar chart directly from city and average Value for money score.

Built one map visual, then duplicated it 4 times, changing the City filter each time to produce the Firenze/Milano/Napoli/Roma/Venezia versions

Measures created included:

- Host Rating
- Superhost Rating
- Host Years
- Superhost Years

Visuals developed:

Pie ChartPie chart showing host and super host average rating.

Donut Chart showing host and super host average years of experience.

Clustered Bar Chart showing the average value of money for each city

Maps showing the average value of money for neighborhoods in each city.

4. Results & Findings

Business Question A

Value for Money, Cleanliness and Accuracy were the strongest drivers of guest satisfaction.

The Key Influencers visual identified **Value for Money as the most influential factor**.

When the average Value for Money score increased by **0.41**, the average Rating Score increased by approximately **0.11**.

The scatter plot also demonstrated a strong positive relationship between Value for Money and Overall Rating.

Business Question B

Higher-priced listings tended to receive slightly higher ratings.

However, the increase was minimal.

Average Listing Price = €173.16

Average Highly Rated Listing Price= €178.50

Difference= €5.34

This suggests tourists do not need to spend substantially more to enjoy highly rated accommodation.

Business Question C

Superhosts achieved higher guest satisfaction.

Average Rating

Superhost = 4.86

Host = 4.64

Average hosting experience:

Superhost = 7.2 years

Host = 7.0 years

This suggests service quality rather than hosting experience distinguishes Superhosts.

Business Question D

Property Size had little influence on guest satisfaction.

Average Ratings

Small = 4.72

Medium = 4.72

Large = 4.72

Very Large = 4.75

Similarly,

Property Types showed only minor differences.

Average Ratings

Entire Home = 4.73

Private Room = 4.70

Hotel Room = 4.66

Shared Room = 4.55

Overall, property characteristics had limited influence on guest satisfaction.

Business Question E

Cities offering the best Value for Money: Napoli = 4.667. Firenze = 4.661

Lowest: Milano = 4.583

5. Dashboard

The dashboard contains multiple interactive report pages including:

Executive KPI Dashboard

Guest Satisfaction Dashboard

Pricing Dashboard

Property Dashboard

Host Dashboard

City & Neighbourhood Dashboard

Interactive slicers allow users to explore the data by city, neighbourhood, property type and pricing category.

6. Limitations & Future Work

Limitations

650 listings contained flagged prices and were excluded from price-based calculations.

Twenty-three listings contained Overall Rating = 0.00 without sufficient documentation explaining their meaning.

The dataset represents one period in time.

The analysis was limited to variables available within the dataset.

Future Work

Future studies could include:

Seasonal demand analysis.

Occupancy rates.

Cancellation behaviour.

7. Recommendations

For Tourists

Prioritise listings with high Value for Money, Cleanliness and Accuracy scores.

Do not rely solely on price when choosing accommodations

Consider highly rated budget and mid-range listings.

Prefer Superhost listings where available.

Consider Napoli and Firenze when seeking the best Value for Money.

Choose accommodation based on guest ratings and reviews rather than property size or accommodation type alone.

8. Appendix

i. Calculated Columns

ii. Price Band

iii. Property Size

iv. Years of Experience

DAX Measures

i. Average Rating

ii. Average Price

iii. Median Price

iv. Total Listings

v. Average Price of Highly Rated Listings

vi. Host Rating

Vii.Superhost Rating

ViiiHost Years

ix. Superhost Years

x. Average Guest Experience Measures

Software Used

i. Microsoft Excel

ii. Microsoft Power BI Desktop